

Live Pipeline

One autonomous agent, one fifteen-minute cycle, seven gates a candidate has to survive before an order is ever built. Two paper accounts ran it side by side, differing in a single parameter. This deck is that system — the deployed one, not the roadmap.

CADENCE

15 min · RTH-gated

UNIVERSE

12 fixed · 6 fixed

DEFAULT

NO_TRADE

MODELS IN PIPELINE

zero

What the agent is, and isn't

One process, one invariant that never bends: **the model proposes, the code disposes**. Everything on the next fourteen slides is that sentence, enforced in code.

SCOPE

A defined-risk, short-dated put-credit book on U.S. equity and sector ETFs via Alpaca — **paper only**. One structure per underlying per cycle, built by rule, not by scanning every combination for the fattest credit.

AUTONOMY

Runs unattended on a 15-minute **launchd** timer through the whole regular session. No human approves a cycle. Every threshold is pre-registered in the account's rulebook and never edited from inside a running cycle.

WHERE A MODEL IS ALLOWED NEAR IT

Exactly one seam, fail-closed: a news classifier that may only *suppress* a candidate on catalyst risk. It cannot create one.

THE INVARIANT

The model never sizes, prices, or authorizes. A malformed or failed model response resolves to **PASS** — never a default open.

MISSING DATA

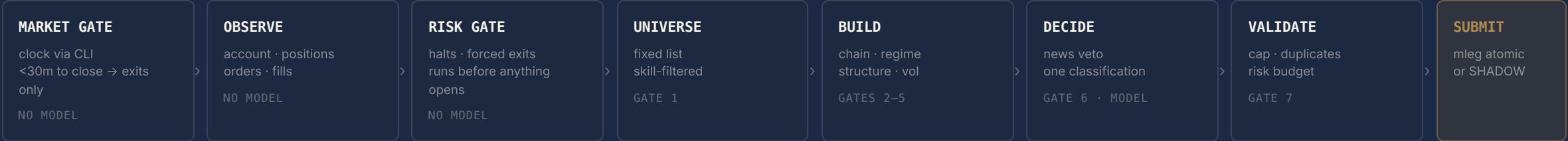
Treated as UNKNOWN, everywhere — never a licence to trade. A stale VIX snapshot does not default to "risk on"; it blocks new short vega book-wide.

WHAT THIS DECK COVERS

The cycle spine, the universe the agent may look at, all seven gates, the two rulebooks that ran, and what the accounts actually did.

Eight hand-offs, every 15 minutes

One tick either dies at the market gate or runs the full chain. Only one box has a model anywhere near it, and it sits after every risk check, not before.



OUTSIDE REGULAR HOURS

`market_gate()` returns before the universe is ever resolved. The journal gets one field — `stage="market_gate"` — and the cycle ends. Zero of seven gates run. On 2026-09-04 that was 44 of 44 ticks on account C.

EXITS RUN BEFORE ENTRIES, ALWAYS

Take-profit, stop, expiry-day close and the flat-by date are evaluated and acted on in the risk gate — before a single candidate is constructed. A cycle that closes a position and opens nothing is the normal, healthy shape.

What the agent is even **allowed** to look at

No scanner, no screener, no ranking model. The universe is a hand-written list in the rulebook, and it is re-filtered against the canon on **every** cycle — a name in the list is a candidate for consideration, never a licence.

01 · THE LIST

`universe.mode = "fixed"`. Account C names twelve underlyings, account B names six. Nothing outside the list is ever priced.

C: SPY QQQ IWM GLD XLF SMH USO SLV TLT IBIT UNG XLV
B: SPY QQQ IWM GLD XLF SMH

02 · CANON FILTER, EVERY CYCLE

`skills.filter_underlyings()` drops on venue and structure grounds — assignment mechanics and decay profile, not a ranking preference.

P5_european_index

V5_vol_etp

P13_crypto_proxy

P13_leveraged_etp

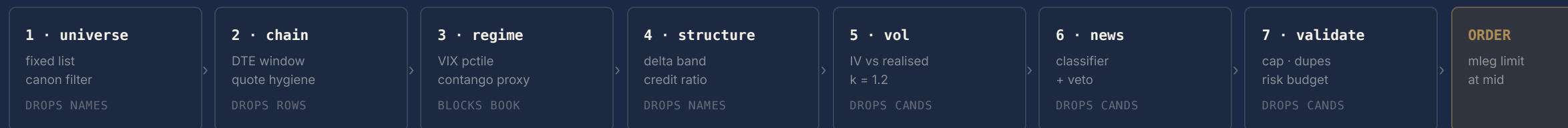
03 · WHAT THAT COSTS, IN FACT

IBIT is dropped on every cycle of account C, by `P13_crypto_proxy`. It is in the rulebook and it has never been tradable by this agent.

The filter is deliberately downstream of the list, so a name added by hand cannot bypass the canon. The cost is that a list can look wider than the book really is — which is exactly what happened here.

Seven gates, **one direction**

Each gate narrows the set or annotates it, then hands off. No gate reorders survivors by expected value, and no gate can re-admit something an earlier one dropped.



THE SHAPE OF A NORMAL DAY

Across 242 cycles on account C, eight structures were submitted and two filled. Six limit orders expired unfilled at the mid and were retired by the pending-order timeout. The gates are not decoration: the overwhelming majority of ticks end in NO_TRADE, and that is the design working rather than the agent failing.

universe

chain

regime

structure

vol

news

validate

universe • Earning the right to be priced

Before any name is priced it has to clear the canon — a venue-and-structure rule that runs on every name, every cycle, no matter what the rulebook says.

MECHANISM

The rulebook's list is the input, not the output. Each name is tested against four structural exclusions, and a drop is recorded with its exact code — a name that vanishes from the book always says why in the journal.

WHY THESE FOUR, AND NOT A SCORE

European-style index options cannot be traded on this venue at all. Volatility ETPs are path-dependent and are never a short-vol warehouse. Leveraged and inverse ETPs carry rebalancing drag that makes them derivatives of a derivative. Crypto proxies duplicate a factor this book does not price.

None of that is a preference the agent could be argued out of on a good day. It is assignment mechanics and product structure, decided once.

IN FORCE

4 structural exclusions

applied to every name on every cycle, downstream of the list

DROP REASONS — REAL CODES

P5_european_index

V5_vol_etp

P13_crypto_proxy

P13_leveraged_etp

A second pass runs later on the finished candidate: VXX is refused as a short-vol warehouse even if it somehow reached that far.

universe chain regime structure vol news validate

The market as it **actually** is

Two gates that do different jobs: **chain** decides which quotes are trustworthy enough to price a leg from; **regime** decides whether short vega is allowed at all, for the whole book at once.

CHAIN · QUOTE HYGIENE

The DTE window is cut first, then clamped again by the book's flat-by date so nothing is opened that cannot be held to expiry. Every remaining row is checked before a structure is built from it. An untrustworthy quote is rejected before the code — or the model — ever sees it.

4–7 DTE · 300s · 8% · \$0.02

min_dte-max_dte · max_quote_age_s · max_spread_pct_of_mid · mid floor

unpriced stale_quote too_cheap spread_too_wide no_expiry_before_flat_by

REGIME · ONE READ FOR THE WHOLE BOOK

Drops no single name. It sets one book-wide switch: may this agent sell vega today at all? Missing or stale data is not permission — it resolves to UNKNOWN and the caller decides conservatively.

pct_{1y} < 0.70 · contango

VIX percentile below the stress decile, and the VIX3M/VIX ratio in contango. Backwardation closes the book to new short vega immediately.

P8_backwardation_vix3m_proxy P8_vix_pct_1y>=0.70 no_vix_snapshot

Stated honestly: the canon asks for VX futures term structure. This venue lists none, so a VIX3M/VIX ratio stands in and is labelled a proxy in the data, not passed off as the real curve.

structure • One shape, built by rule

There is no menu the agent picks from and no family selection. The book sells one structure — a put credit vertical — and this gate either constructs it or returns nothing.

SHORT LEG → LONG LEG → SANITY

1. Take every put whose absolute delta sits inside the band, then pick the one closest to target. **2.** Place the long leg one width below it — or, when that strike is not listed, at the *nearest listed strike to that distance*. **3.** Reject unless the credit-to-width ratio lands inside its corridor. **4.** Size from the account's risk budget; a size of zero is a drop.

WHAT THE NEAREST-STRIKE FALLBACK ACTUALLY DID

Nominal width is 5.0. Live, the fallback produced widths of **1.0, 2.0, 3.0, 4.0, 5.0 and 6.0** across the fifteen spreads the two accounts submitted — because the ETF strike grids are not uniform at these tenors. The book is therefore less homogeneous than the single number in the rulebook suggests, and the journal records the width actually built.

THRESHOLDS IN FORCE

$$|\Delta| = 0.15 \pm 0.08$$

target_delta · delta_tolerance

width **5.0** · nearest listed

credit/width **0.08–0.60**

min_credit_ratio_vertical · max_credit_ratio

DECLINES, HONESTLY LABELLED

When the builder finds nothing it journals **no_structure** and does *not* invent a reason — it does not report which of its own four conditions bound. That is a stated limit of the trace, not a gap someone forgot to fill.

vol · Sell only what is rich

The single thesis of the book, applied to the finished structure rather than to a name: implied volatility on the leg being sold has to exceed recent realised by a margin.

MECHANISM

Realised volatility is computed over a 21-session window from the same OHLCV helper the rest of the agent uses. The implied side is the **short leg's own IV**, not an at-the-money proxy — the number belonging to the contract actually being sold.

The ratio must clear **1.2**. Below it the premium does not compensate for the risk being taken, and the candidate is dropped rather than resized.

NO DATA IS NOT A PASS

If realised volatility cannot be computed at all, richness cannot be verified, and the candidate is dropped. Missing evidence never resolves in favour of trading.

CONSTANT

$$IV_{\text{short}} / RV_{21} \geq 1.2$$

$$\text{min_iv_rv_ratio} \cdot \text{rv_window} = 21 \cdot \text{vol_gate_iv} = \text{short_leg}$$

WHY THIS ONE IS LOAD-BEARING

Live and backtest share this gate through the same code path. If the two ever disagreed about what "rich" means, the backtest would stop describing the agent — so the live path and **backtest/strategy.py** apply the identical rule rather than two implementations that happen to match today.

news • The one place a model touches this

One classification call per cycle. It returns three enums. Deterministic code turns those into ALLOW or PASS per underlying. **The model cannot create, size or price a candidate** — by this point every candidate already exists and has passed five gates.

THE CONTRACT, IN FULL

FIELD	VALUES
catalyst_scope	NONE · SPY_ONLY · QQQ_ONLY · BOTH · UNKNOWN
jump_severity	NONE · MINOR · MAJOR
news_clarity	CLEAR · UNCLEAR

It sees the filtered headlines, the DTE, and the short strike for a distance test. It does not see thresholds, sizing, P&L, the legs, or the account. Schema-constrained decoding; the response is validated against the enum set and rejected if it drifts.

HOW THE ENUMS BECOME A DECISION

UNCLEAR or UNKNOWN → PASS. Uncertainty resolves to no trade.
Scope touches the name and severity is MAJOR → PASS.
MINOR → logged, allowed. A sub-1% expected gap cannot reach a 15-delta short strike.
Quiet tape → ALLOW.

EVERY FAILURE MODE, ONE ANSWER

No API key · transport error · refusal · malformed JSON · an enum outside the schema · an empty news feed — all of them resolve to the same conservative context, and that context resolves to PASS. The agent degrades to **trading nothing**, never to trading without the check.

universe

chain

regime

structure

vol

news

validate

validate • The last word before an order

Everything upstream is re-checked against live account state, because the account can have moved since the cycle began. A candidate that survived six gates still has to fit.

CAPACITY

3 concurrent

structures, not legs. One open four-leg position is one position — a bug worth naming, because counting legs would let a single condor exhaust the whole book.

1 per name + expiry

max_per_name. A second spread on the same underlying and expiry is dropped, not stacked.

BUDGET

2% equity per trade

Max loss is width less credit, known exactly before entry, so size is arithmetic rather than estimate. Capped at 20 contracts.

vol-target 0.25

Scales the budget 0.25×–2.0× from trailing marked-equity volatility.

HALTS, AND WHO CLEARS THEM

–4% day · –10% drawdown

Both measured on the **marked** curve, because a realised-cash curve cannot see a drawdown that has not been closed yet.

A breach stops new entries and **requires a human to clear the state file**. The agent will not restart itself. This is the one place the design refuses to be autonomous.

What an order actually is

Both legs go as one package. A spread is one position with one price, not two orders that might half-fill and leave a naked short behind.

SUBMIT

Multi-leg mleg, limit at mid. Never a market order — the chain feed is indicative, modelled rather than a real book, and a market order against it would be reckless.

Every order carries a client_order_id. Alpaca rejects duplicates, which is what makes a retry safe instead of a double-fill.

Unfilled limits are retired after the pending-order timeout, so a lost order id cannot occupy the position cap for the rest of the session. Six of account C's eight submissions ended exactly this way.

JOURNAL — THE PART THAT MAKES ANY OF THIS CHECKABLE

Every cycle writes two files: a JSONL line carrying every input, gate result, model response and order, and a human-readable analysis log of the same cycle.

Both ship in this repository for both accounts — `agent/profiles/*/decisions/`. They are the evidence for every claim on the previous ten slides, including the cycles where the agent looked at the market and decided to do nothing, which is most of them.

RECONCILIATION, EVERY CYCLE

Before judging anything, the agent compares what it believes it holds against what the broker reports. An unregistered leg or an equity position halts it — assignment is handled by stopping, not by guessing.

Two books, one agent

Accounts B and C run the **same agent_loop.py**. What differs between them is a JSON rulebook, and those two rulebooks differ in exactly three things that matter. Everything else is byte-identical — which is what makes the pair a controlled comparison instead of two unrelated bets.

	ACCOUNT C	ACCOUNT B
rulebook	putcr-core6-d47	putcr-core6
account	PA3UU4TX8Y3K	PA3ZNXHF8ID1
min_dte → max_dte	4 → 7	1 → 7
underlyings	12	6
flat_by_date	2026-09-18	2026-09-03
timer	15 min	30 min
target_delta ± tol	0.15 ± 0.08	0.15 ± 0.08
width · credit ratio	5.0 · 0.08–0.60	5.0 · 0.08–0.60
min_iv_rv_ratio	1.2	1.2
risk · concurrent · per name	2% · 3 · 1	2% · 3 · 1
take profit · stop	50% · 2×	50% · 2×
halts	−4% · −10%	−4% · −10%

WHY A RULEBOOK AND NOT A BRANCH

A profile binds a rulebook, a journal directory and an account together. **assert_account** reads the broker before any order and **refuses to start** if the credentials reach an account other than the one the book names. Two books cannot silently share one account, which is what makes their journals comparable.

THE VARIABLE UNDER TEST

Tenor. C sells 4–7 days out, B sells 1–7. Same strikes rule, same richness gate, same sizing, same exits. Any difference in outcome is the tenor or it is noise — there is no third explanation hiding in the configuration.

READING THE REASONING

Every rulebook carries **_why** fields beside its numbers. The argument for a threshold lives next to the threshold, so it cannot be changed without seeing what it was for.

Live, and measured

Both accounts started at \$100,000.00 and both are up. The backtest disagrees with the ranking, and that disagreement is the most useful thing on this slide.

LIVE · PAPER, 1–4 SEPTEMBER 2026

	C · PA3UU4TX8Y3K	B · PA3ZNXHF8ID1
equity	\$100,342.80	\$100,277.20
return	+0.343%	+0.277%
cycles	242	131
spreads submitted	8	7
filled	2	7
unfilled at the limit	6	0
open now	1 · SPY 749/754P	0 · flat

BACKTEST · 2024-01-22 → 2026-08-26 · 1,661,101 OPTION BARS

FRICTION	C · DTE 4–7	B · DTE 1–7
1% per leg	+18.23% · 0.86	+16.62% · 0.68
3% per leg	–3.79% · –0.14	+11.51% · 0.51
6% per leg	–25.11% · –1.21	+2.75% · 0.16
trades	312	233
win rate	76.9%	87.1%

return · Sharpe, net of modelled fees. SPY buy-and-hold over the same window: +58.19%, Sharpe 1.21 — neither book beats it.

WHAT THAT CONTRAST MEANS, SAID PLAINLY

C leads on live P&L. B is the book that survives the friction sweep. Three days on two filled spreads is noise, not evidence; two and a half years on 312 trades is evidence, and it says C's tenor does not clear realistic execution cost. A deck that showed only the live column would be showing the weaker of the two numbers it has. There is no historical option bid/ask on this venue, which is why the sweep exists at all — a book that is positive only at 1% is not a book.

The agent, summarized

Four facts that do not change with market conditions, a flattering backtest, or a persuasive headline.

ZERO MODELS BUILD A CANDIDATE

Gates 1 through 5 are code: thresholds from the rulebook, mathematics from `canon.py`. No model sits between a name entering the funnel and a structure coming out of it.

PAPER ONLY, STRUCTURALLY

No code path in this repository constructs `--live`. It is not a setting to be careful with; the live path does not exist. A test asserts it.

ONE LIVE MODEL CALL, AND IT CAN ONLY SUBTRACT

The news classifier runs once per cycle, after five gates. It may suppress an open on catalyst risk. It never proposes, sizes or prices a structure, and every failure of it resolves to PASS.

HALTS NEED A HUMAN

A -4% day or -10% drawdown stops new entries and requires someone to clear the state file by hand. The one thing this agent is not allowed to decide is whether it should keep going after it has been wrong.

Autonomy with a leash it **can't chew through**.

Every spread this agent ever submitted had already survived seven gates it cannot argue with. The one place a model's judgment enters is a veto that can only ever say no — and everything it would have to get past to say yes was decided in code, in advance, and written down beside the number.

THE DEPLOYED SOURCE, IN READING ORDER

agent_loop.py	cycle · gates 1–5 · validate · mleg submit
veto.py	classification schema · deterministic veto
skills.py	P5 / V5 / P13 filters · VIX regime gate
canon.py	HAR forecast · MAE · VIX context
structures.py	structure-level P&L · reconciliation
preflight.py	23 live-dependency checks
profiles/*/rules.json	every threshold in this deck
backtest/	engine · strategy · fills · store · run

VERIFICATION

Every threshold, drop-reason code and gate on these slides was read from the deployed source on **2026-09-04**, and a script re-checks each number against the rulebooks and run summaries at build time. Nothing here is paraphrase or an illustrative figure.

REPOSITORY

alpaca-options-agent-hackathon
lablab.ai × Alpaca · paper-only
make restore && **make test** && **make sweep** reproduces every number on slide 13 from a clean clone, offline.